

Derivation, Parse tree, Ambiguity

Qa)

$$S \rightarrow ASB|SS|TSS|SAS|A$$

$$A \rightarrow ASS|BS|B$$

$$B \rightarrow 00|11|01|1$$

a) left derivation of 00010111

$$S \rightarrow \underline{S}AS$$

$$\rightarrow \underline{SS}AS$$

$$\rightarrow \underline{AS}AS$$

$$\rightarrow \underline{BS}AS$$

$$\rightarrow 00\underline{S}AS$$

$$\rightarrow 00\underline{A}AS$$

$$\rightarrow 00\underline{B}AS$$

$$\rightarrow 0001\underline{A}S$$

$$\rightarrow 0001\underline{B}S$$

$$\rightarrow 000101\underline{S}$$

$$\rightarrow 000101\underline{A}$$

$$\rightarrow 000101\underline{B}$$

$$\rightarrow 00010111$$

[underlined variables were derived in the next substitution]

No need to show in the answer script. If you want you may.

[Multiple derivation is possible for this grammar and the string]

b) right derivation for 00010111

$$S \rightarrow SAS\underline{}$$

$$\rightarrow SAA\underline{}$$

$$\rightarrow SA\underline{B}$$

$$\rightarrow SA11\underline{}$$

$$\rightarrow S\underline{B}11$$

$$\rightarrow S\underline{0}111$$

$$\rightarrow S\underline{S}0111$$

$$\rightarrow S\underline{A}0111$$

$$\rightarrow S\underline{B}0111$$

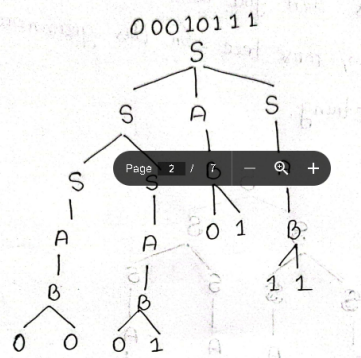
$$\rightarrow S\underline{0}0111$$

$$\rightarrow A010111$$

$$\rightarrow B010111$$

$$\rightarrow 00010111$$

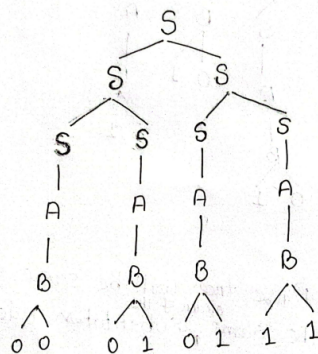
c) Parse Tree for the derivation given in (a)



d) Parse Tree for the derivation given in (b)

Same as 'c'

If you were asked to show that the Grammar is ambiguous, then you have to show another left parse tree for this grammar on the given string.



[There are multiple possible parse tree apart from this.]

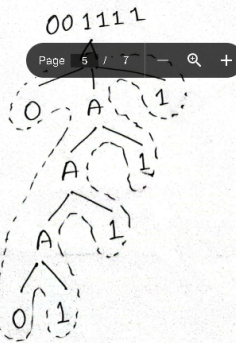
Qb) $A \rightarrow A1 \mid 0A1 \mid 01$

a) leftmost derivation for 001111

$A \rightarrow 0A1$
 $\rightarrow 0A11$

→ 0 A 1 1 1
→ 0 0 1 1 1 1

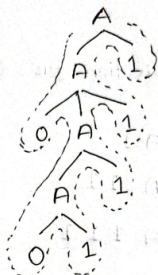
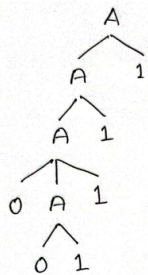
b) Parse tree for the derivation in (a):



[..... not necessary]

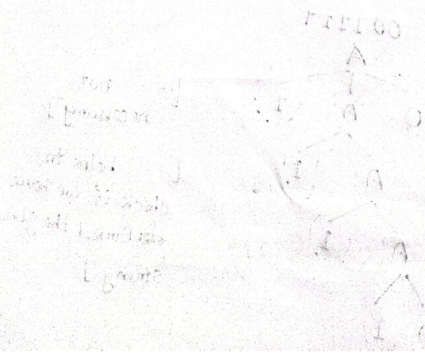
[..... helps to check if we have der. parsed the given string]

c) Two more Parse trees



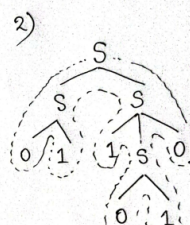
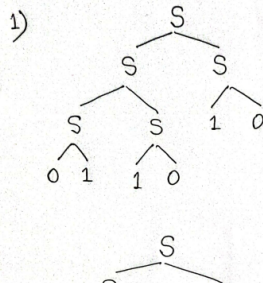
[..... not necessary]

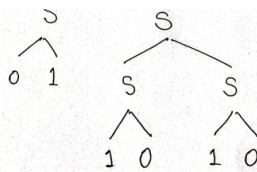
d) 000111 or 011111:



9c) $S \rightarrow 0S1 \mid 1S0 \mid SS \mid 01 \mid 10$

a) Two different parse tree for 011010.





b) 000111 or 111000 or 001110 or 011100
or 110001 or 100011.

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